

Echoes of the Ancestors

Visayas Island Group Focus

1. The Core Feature

What It Does

- Scan an instrument using the device camera, or upload a photograph
- Gemini AI identifies the instrument — name, region, cultural context
- The identified instrument becomes playable in an interactive rhythm game
- The player will have the ability to unlock areas all across Visayas in the span of the story.
- End game content includes dynamically created AI students or real students that the players will teach to solidify their knowledge about the learnings from the game.

The Critical Design Problem

The upload function creates a shortcut that breaks any collection-based game loop. A player can search 'Tultugan' on Google, upload the image, and complete their collection in minutes — bypassing all engagement mechanics.

Recommended Solution: Two-Mode System

Upload = Learn Mode. Identifies the instrument, plays audio, shows cultural context. No collection progress. No XP.

Camera Scan = Discovery Mode. This is the game. Scanning a real instrument unlocks it in the collection, earns XP, and fills the regional map.

This is the pitch to judges: 'You can look anything up — but to own it, you have to find it.'

This framing connects the app to cultural preservation: students are incentivized to visit museums, attend fiestas, and engage with community instrument collections — not just browse Google Images.

2. Gamification Philosophy

The Mechanic Hierarchy

Layer	Mechanic	Purpose
Core Action	Scan (camera)	Cheat-proof discovery mechanic
Skill Loop	Rhythm game + Guess-The-Instrument	Rewards mastery after discovery
Progress System	XP + Levels + Regional map	Makes progress visible and satisfying
Retention System	Streaks + Daily challenges	Brings players back daily
Social Layer	Leaderboard + Classroom mode	Competitive motivation
Completionism	Collection + Badges	Drives finishing behavior

3. Full Player Journey

Day 1 — The Call of the Islands

The player opens the app for the first time. No tutorial dump. Just a map of the Visayas, most regions in fog.

- Onboarding screen: "A great cultural journey begins. Point your camera at a traditional instrument to start."
- First Quest auto-unlocks: Become a Music Apprentice — discover 1 instrument. Player scans their first instrument using the camera. AI identifies it. Audio plays. Cultural story appears. +5 XP.
- Rhythm game launches immediately — player plays the instrument for the first time.
- Mini-tutorial triggers: Guess-The-Instrument quiz round fires after the rhythm game.
- Correct answer: +10 XP, streak begins. Level-up animation: Level 1 — Apprentice. Session ends on a cliffhanger notification: "2 more instruments before Western Visayas fully opens..."

Note: Players who use Upload mode can still access the rhythm game and quiz, but earn no XP and no collection progress.

Day 1 Rewards

Badge: First Discovery · +15 XP · Streak Day 1 · Level 1 unlocked

Days 2–4 — Completing the First Expedition

The habit-building window. Daily challenges keep the player returning. Each session has one clear micro-goal.

- Day 2 challenge: Listen to 2 audio samples. Reward: +50 XP. Streak protected.
- Day 3: First story scenario triggers — "A barrio fiesta in Iloilo is starting. Which instrument leads the procession?" Tests knowledge from Day 1.
- Day 4: Final quest objective — complete the Western Visayas quiz (5 questions, 80% passing score). Difficulty scales to what the player has already scanned.
- Quest complete animation fires. **The Panay Tribesman** badge is earned for discovering the core indigenous instruments. Map fog clears — Central Visayas glows.
- *Hidden Event*: If the player scanned the Pasiyak during this window, the secret **Water Whistler** achievement pops. Leaderboard appears for the first time. Player sees classmates ranked. Social hook is planted.

Days 2–4 Rewards

Badges: The Panay Tribesman & Water Whistler (Hidden) · Level 2 unlocked · 4-day streak · Leaderboard access

Days 5–14 — The Central Visayas Arc

Difficulty ramps. New mechanics are introduced gradually so the player is never overwhelmed.

- Harder Guess-The-Instrument rounds — similar-sounding instruments from the same family appear together (like the Bandurria and Laud).
- Collection system introduced: "3/5 Central Visayas string instruments discovered. 2 locked cards remain." Completionist loop activates.
- Upon collecting enough non-bamboo items across regions, **The Wood Carver** badge unlocks. Streak Shield item unlocks at Day 7 — earned by 80%+ quiz accuracy. Protects streak if player misses a day.
- First Cultural Keeper story scenario: "A community is about to lose knowledge of their instrument. Help preserve it."
- Day 14 — Level 3 milestone. Village Musician title unlocked. Completing the string collection triggers the **Rondalla Master** badge. Weekly classroom leaderboard resets.

Days 5–14 Rewards

Badges: Rondalla Master & The Wood Carver · Streak Shield item · Village Musician title · Weekly leaderboard reset

Days 15–30 — Eastern Visayas and Negros Region

By now the player has a strong habit. Difficulty peaks. Mastery mode and end-game loops are previewed.

- Timed Guess-The-Instrument rounds — 10 seconds per answer. Streak multiplier activates for correct runs.
- Eastern Visayas introduces the rarest courtship instruments (Lantoy, Korlong) with the richest cultural context. Scanning them all unlocks the **Eastern Courtship** badge.
- Day 20: Cultural Keeper level unlocked. Bonus lore cards and extended audio recordings become available.
- Day 25: First Master Challenge — a 20-question mixed-region quiz testing all 14 items.
- Day 30: All expeditions complete. Collecting the final instrument triggers the ultimate completionist badge: **Visayan Virtuoso**. Master Instrumentalist title granted. Mastery Mode unlocks.

Days 15–30 Rewards

Badges: Eastern Courtship & Visayan Virtuoso · Master Instrumentalist title · 100% collection · Mastery Mode

Day 30+ — The End-Game Loop (Mastery Mode)

The answer to "what happens when you've learned everything?" The game never truly ends.

- **Weekly Rotating Challenges** — A new themed challenge drops each week (e.g. "Visayas String Week" — only string instruments, harder questions). No new content needed, just harder skill tests.
- **Speed Trials** — Recognize 10 instruments in under 90 seconds. Leaderboard resets weekly so new players can compete.
- **Teach Mode** — Player "teaches" a virtual student by identifying and explaining instruments. Same multiple-choice format, but framed as the player being the expert.
- **Prestige System** — At max level, player can reset XP (keeping all badges) for a star marker on their profile. Hardcore players chase this.

4. Progression Arc

Level Structure

Level	Title	What Unlocks
Level 1 — 0 XP	Apprentice	Western Visayas instruments (Panay), basic instrument cards, 3 daily challenges
Level 2 — 100 XP	Village Musician	Central Visayas instruments (Cebu/Bohol), Streak Shield item, leaderboard access
Level 3 — 250 XP	Cultural Keeper	Eastern Visayas instruments (Samar/Leyte), bonus lore cards, timed round mode
Level 4 — 500 XP	Regional Expert	Negros Island Region (Crossover zone), extended audio clips, Speed Trial access
Level 5 — 900 XP	Master Instrumentalist	All Visayas content, Mastery Mode, Prestige system, Teach Mode

XP Sources

Action	XP Earned + Notes
View instrument card	+5 XP — first view only, no farming
Listen to audio sample	+5 XP — first listen only
Complete quiz	+20 XP — any score
Perfect quiz score	+30 XP — replaces the +20
Correct mini-game answer	+10 XP — per round
Daily challenge complete	+50 XP — streak multiplier applies
7-day streak milestone	+100 XP — bonus, once per streak
Expedition complete	+150 XP — one-time per region

5. Fleshed-Out Mechanics

5.1 Guess-The-Instrument Mini-Game

Three difficulty tiers based on player level. Each correct answer earns +10 XP. A streak multiplier fires at 3 and 5 correct answers in a row.

Tier	Who Sees It	How Hard
Tier 1	Apprentice (Level 1)	4 choices from different instrument families
Tier 2	Village Musician (Level 2+)	4 choices from the same family (e.g. all string)
Tier 3	Cultural Keeper (Level 3+)	4 choices from the same region with similar timbre

Example — Tier 3 Round

[Sound plays — a deep, echoing strike against hollow bamboo]

A) Litguit B) Tugo C) Tultugan D) Buktot

Correct: C — Tultugan. "The massive bamboo Tultugan provides the booming foundational beat for traditional Ilonggo festivals."

5.2 Streaks — Solving the Content Cliff

The classic streak problem: players run out of new instruments to discover. A streak tied to 'learn something new' dies when content runs out.

Solution: decouple streaks from new content discovery. A streak tracks 'engaged with the app today,' not 'found something new.' Daily challenges rotate through speed trials, review quizzes, and story scenarios — no new instruments required.

Streak Protection	How It Works
Streak Shield	Earned by 80%+ quiz accuracy. Protects streak for one missed day. Max 1 per week.
Content rotation	After all instruments are found, daily challenges use review, dynamically created story scenarios and speed modes — no new content needed.
Weekly reset	Classroom leaderboard resets every Monday. A dedicated week can beat a month-long lead.

Example — Day 35 Challenge (Post-Completion)

Match 5 Visayas instruments to their cultural context in under 60 seconds.

No new content needed. The challenge is the skill test, not the discovery.

5.3 Story Scenarios

Each region has 3 scenarios that unlock as the player discovers instruments there. Format: a short scene (2–3 sentences), a decision point, then a consequence. Wrong answers don't fail the player — they trigger a cultural explanation. This keeps learning exploratory, not punishing.

Example — Visayas Scenario

An elder at a harvest festival in Iloilo asks you to provide a loud, heavy beat for the tribal dancers. Which indigenous instrument do you pick up?

A) Bandurria B) Kulintang C) Buktot D) Tultugan

Correct Answer: (D) Tultugan

Wrong answer (B): The Kulintang is a brass gong ensemble from Mindanao! To keep the beat at a Visayan festival, you need the massive bamboo Tultugan drums. +5 XP still awarded.

5.4 Leaderboard — Anti-Frustration Design

Three separate boards prevent new players from feeling permanently behind:

- Classroom board — your class only. Most motivating for school users.
- Friends board — connected accounts. Social accountability loop.
- All-Time board — global. Shows prestige stars, not raw XP. Reachable through dedication, not just time played.

Weekly boards reset every Monday. A single dedicated week of play can beat a month-long lead. The board always shows the XP gap to the next rank — not just your position.

Example Display

This week in your class:

1. Maria — 480 XP
2. Juan — 410 XP
3. You — 390 XP ← 20 XP to reach 2nd place

5.5 Collection System

Instrument cards show a silhouette and region hint when locked. Players can see exactly how many remain ('4 instruments still hidden in Mindanao'). Completing a region's full collection triggers a special animation and grants the region's rare badge — a much larger reward than individual unlocks.

Locked Card Example

[Silhouette visible] — Region: Western Visayas

A hunchbacked, four-stringed native guitar crafted from a dried coconut husk.

Unlock by completing the Panay Island expedition quiz.

5.6 Core Gameplay Mechanics

Rhythm & Scoring System

The core moment-to-moment gameplay is driven by a precision rhythm engine. Players must perform inputs (plucks, strikes, or breaths) in synchronization with the procedurally generated ancestral beat. Hit judgements are calculated based on strict millisecond windows:

Example

Perfect (Tadhana): $\pm 45\text{ms}$ window. Awards 100 points and triggers a golden particle burst.

Good (Ganda): $\pm 90\text{ms}$ window. Awards 50 points and sustains the combo.

Miss (Sablay): $> 90\text{ms}$ window. Awards 0 points, triggers a visual UI glitch, and breaks the current streak.

Combo & Flow Multiplier

The game utilizes a tiered multiplier system designed to reward sustained mastery. The multiplier snaps to higher tiers as the player maintains an unbroken combo:

Example

5+ Combo: 2× Multiplier

10+ Combo: 3× Multiplier

20+ Combo: 4× Multiplier (Max Flow State)

Any "Miss" immediately resets the combo to zero, dropping the multiplier back to 1×.

Instrument Control Schemes

The UI dynamically rebuilds itself based on the scanned instrument, offering highly realistic tactile feedback.

- **String Mechanics (e.g., Kudyapi):** The interface renders a vertical fretboard. Players interact by tapping, holding, or dynamically swiping their fingers across the screen to "pluck" the virtual strings. Desktop players can mirror this using keyboard number mapping.
- **Wind Mechanics (e.g., Tongali):** Designed for maximum realism, players do not simply "tap" holes. They must use a Sustained Input (Hold-to-Blow). Players either physically blow into their device's microphone (tracked via volume threshold algorithms) or hold an on-screen breath trigger. While blowing, players hold specific combinations of finger nodes to realistically modulate the pitch. Releasing the breath trigger immediately cuts the sound.

Real-Time Acoustic Engine

Rather than relying on static, pre-recorded MP3s, the game uses a dynamic acoustic synthesizer to recreate physical sound behaviors locally:

- **Plucked Strings:** Utilizes physical modeling (Karplus-Strong algorithms) to simulate the bright, woody pluck and organic acoustic decay of a real lute.
- **Struck Bronze (Kulintang):** Employs Frequency Modulation (FM) synthesis with inharmonic overtone ratios to perfectly recreate the metallic shimmer, sharp attack, and long resonant decay of heavy brass gongs.

5.7 AI & Procedural Systems

The 5-Step Agentic Pipeline

To allow infinite scalability, the game uses an autonomous multi-agent pipeline to turn any uploaded image or camera scan into a playable level:

- **Vision & Cultural ID:** The AI analyzes the camera feed, identifying the instrument, its origin, and its category (String/Wind/Percussion).
- **Acoustic Modeling:** Simultaneously, the system extracts the instrument's fundamental physics (decay time, timbre, native scale).
- **Input Mapping:** The engine translates the physical mechanics of the instrument into the ideal touchscreen control layout.
- **Cultural Guardrail:** An automated checker cross-references the generated data against authentic Philippine musical structures, correcting any modern 12-tone inaccuracies to match indigenous microtonal intervals.
- **Synthesis:** The data is fused into an `ActiveInstrumentProfile`, instantly generating the rhythm level without manual developer input.

Dynamic Story Scenarios (Procedural Narrative)

To solve the "content cliff" of traditional educational games, the game employs an LLM to generate context-aware scenarios. The AI evaluates the player's current region, their unlocked instruments, and their recent performance to generate dynamic cultural choices (e.g., "An elder asks you to lead the harvest procession. Which of your unlocked instruments do you use?"). This ensures the narrative is always personalized and endlessly replayable.

End-Game "Teach Mode" Persona Engine

Upon reaching max level, the AI reverses the standard educational dynamic. Instead of lecturing the player, the LLM assumes the role of a virtual, flawed student with a distinct personality (e.g., an over-eager beginner, or a traditionalist afraid of mistakes). The player must use the knowledge they have accumulated to teach and correct the AI's techniques, cementing their own mastery.

6. The Gameplay Loop

A single session flows through the following chain. Each step creates a reason to do the next one.

Step	What Happens
1. Open Daily Challenge	Streak protected, +50 XP. Clears immediately — low barrier to start.
2. Explore Region Map	Player taps a fogged or undiscovered area. Curiosity is the driver.
3. Scan / Discover Instrument	Camera scan or (learn-mode) upload. Audio plays, cultural story loads. +5 XP.
4. Rhythm Game	The player must play the generated rhythm track (plucking/blowing) to "synchronize" with the instrument. Achieving an 80% score unlocks the lore.
4. Play Mini-Game	Guess-The-Instrument round fires immediately. Streak multiplier applies.
5. Story Scenario (if triggered)	Cultural decision point. Bonus lore card if answered correctly.
6. XP + Level Progress	Progress bar updates. Level-up animation fires if threshold is hit.
7. Check Leaderboard / Collection	Social comparison. Locked card curiosity pulls toward next session.
8. Expedition Complete?	Province Badge awarded. Next region unlocks on the map.
9. Return Tomorrow	Push notification: '2 instruments left in Visayas...' Session ends on a hook.

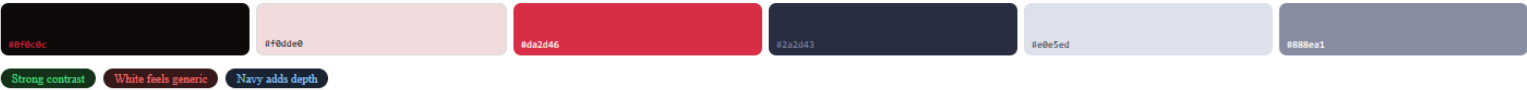
7. Art & UI Direction

7.1 Crimson & Navy - Malong / Terno pattern

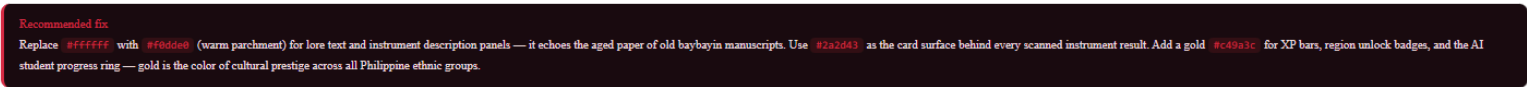
The near-black base paired with crimson mirrors the bold geometry of Malong and T'nalak textile patterns, colors that Filipino weavers have used for centuries to mark significance and ceremony. Our UI, crimson becomes the pulse of the game: it fires on every successful instrument scan, lights up the rhythm beat indicators, and marks newly unlocked Philippine regions on the map. The navy works perfectly as the surface for instrument info cards — dark enough to feel immersive, distinct enough to separate content layers. The slate-gray carries all readable body text, giving the UI a cool, controlled contrast against the warm crimson exactly how modern games like Valorant separate action from information.

- **Near Black (#0f0c0c):** An extremely dark, almost pitch-black shade with a very faint warm/reddish undertone. Used for dark-mode backgrounds or deep text.
- **Pale Dusty Pink (#f0dde0):** A soft, very light pink mixed with a touch of gray, giving it a muted, pastel appearance.
- **Crimson / Cherry Red (#da2d46):** A bold, vibrant red with a slight cool/pinkish tint. This is highly saturated and often acts as a primary accent or alert color.
- **Dark Slate / Gunmetal (#2a2d43):** A deep, dark charcoal mixed with navy blue.
- **Light Cool Gray (#e0e5ed):** A very pale, off-white gray with a subtle bluish tint. Used for light-mode backgrounds or subtle UI borders.
- **Slate Gray (#888ea1):** A medium, muted gray with distinct cool blue undertones.

PAGE 1 Crimson & Navy — Malong / Terno pattern



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7.2 Procedural Weaving UI (The T'nalak Engine)

As the player successfully sustains a combo in the rhythm game, the background of the note highway procedurally weaves a glowing, geometric T'nalak or Inabel textile pattern. This translates the player's accuracy (weaveProgress float) directly into the creation of a visual cultural artifact.

7.3 Dynamic Visual Feedback

The UI responds physically to the player's performance:

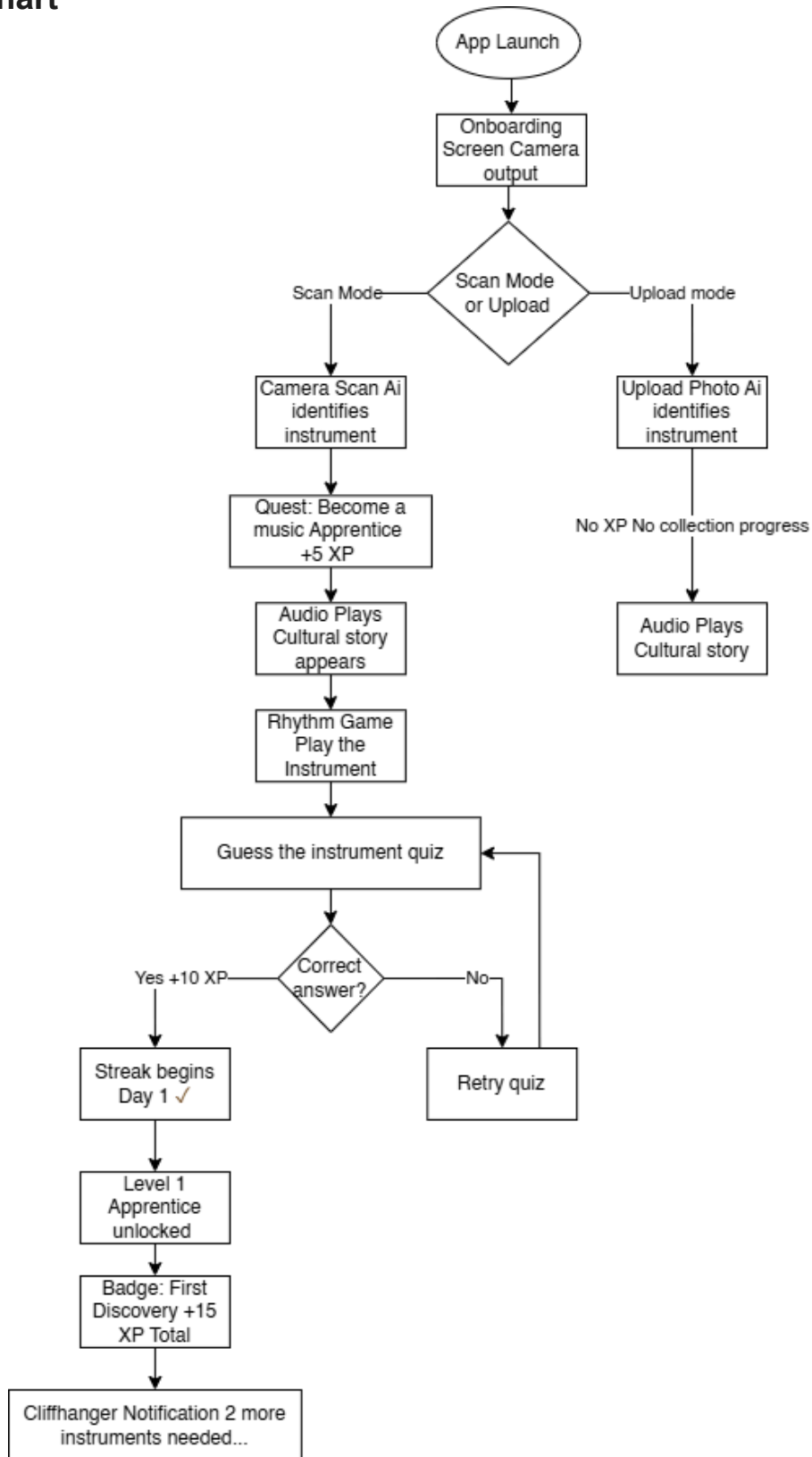
- **Pluck Physics:** Striking a note causes the visual string to vibrate violently on the X-axis, scaling up slightly. Instead of a generic glow, the string emits a vibrant **Crimson (#da2d46)** energy burst that slowly fades into a soft **Pale Dusty Pink (#f0dde0)** as the acoustic sound decays against the **Near Black (#0f0c0c)** background.
- **Judgement Pops:** Accuracy feedback (Tadhana/Ganda) violently scales into the center of the screen, utilizing crisp **Light Cool Gray (#e0e5ed)** typography with subtle **Slate Gray (#888ea1)** drop shadows for depth, before dissolving seamlessly into the interface.
- **Glitch State:** A missed note triggers an aggressive screen-tear glitch effect, flashing stark **Crimson (#da2d46)** and **Light Cool Gray (#e0e5ed)** artifacts. This brief visual chaos cuts the music and unravels the woven **Dark Slate (#2a2d43)** and **Near Black (#0f0c0c)** background elements to signal a jarring break in the ancestral connection.

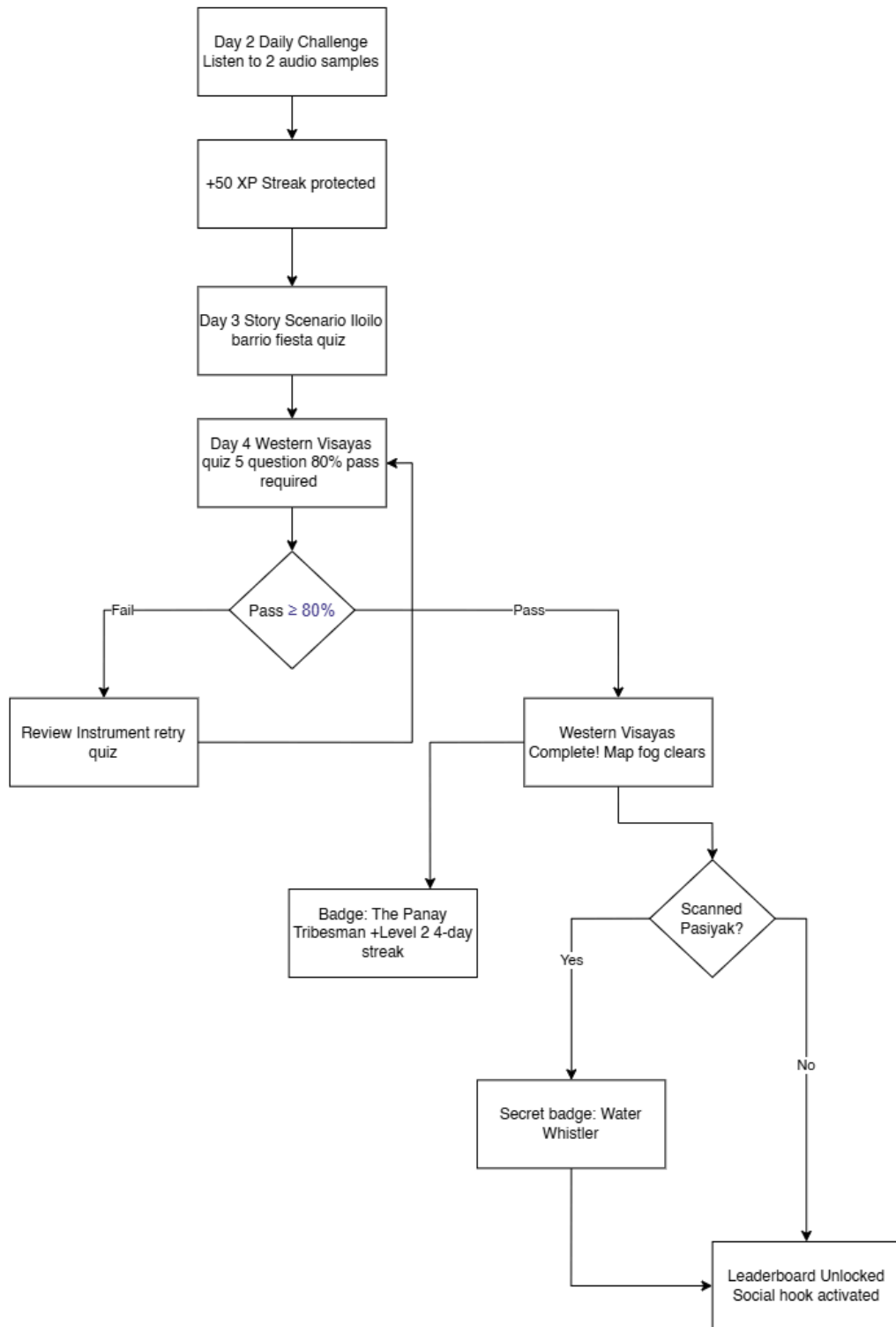
7.4 Instrument Discovery Card (Interactive)

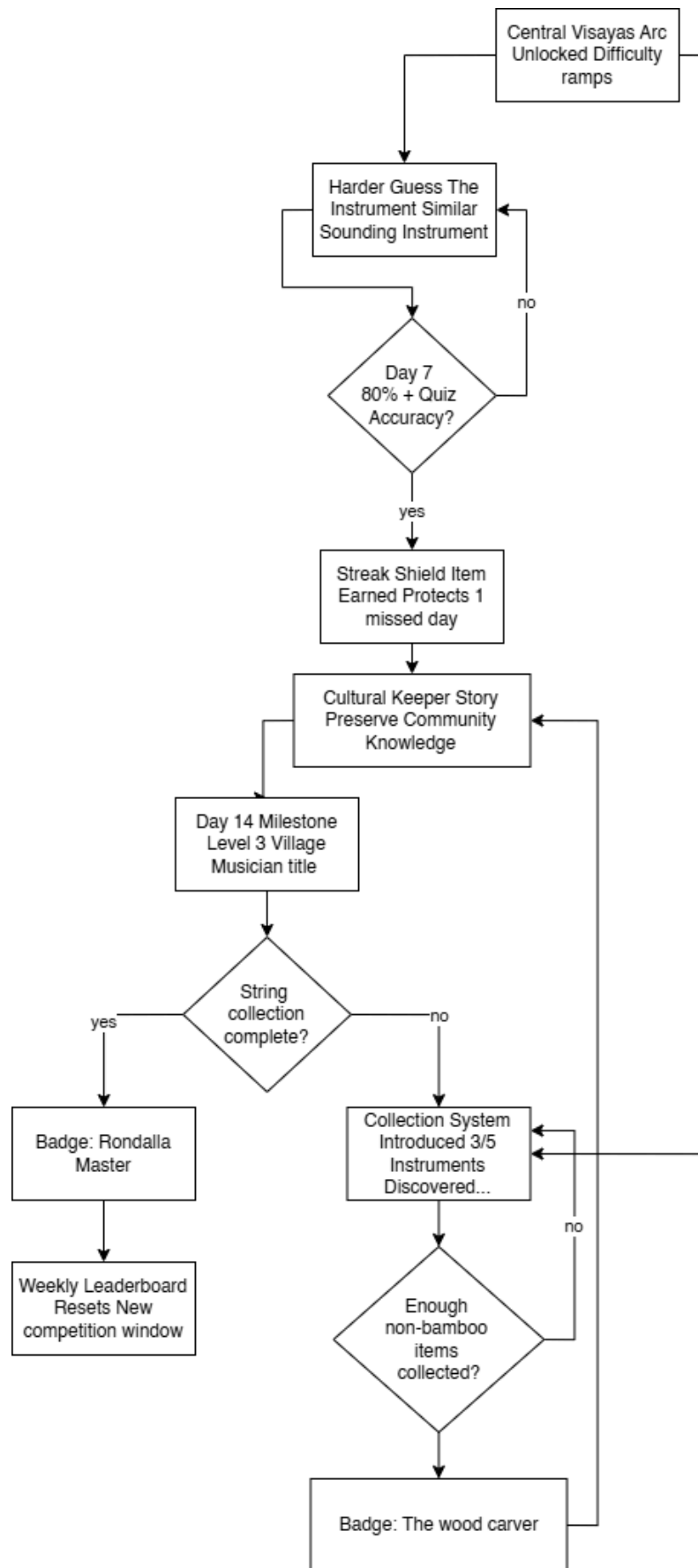
Upon successfully catching or unlocking a new instrument, the gameplay briefly pauses to present a high-fidelity, interactive 3D presentation card.

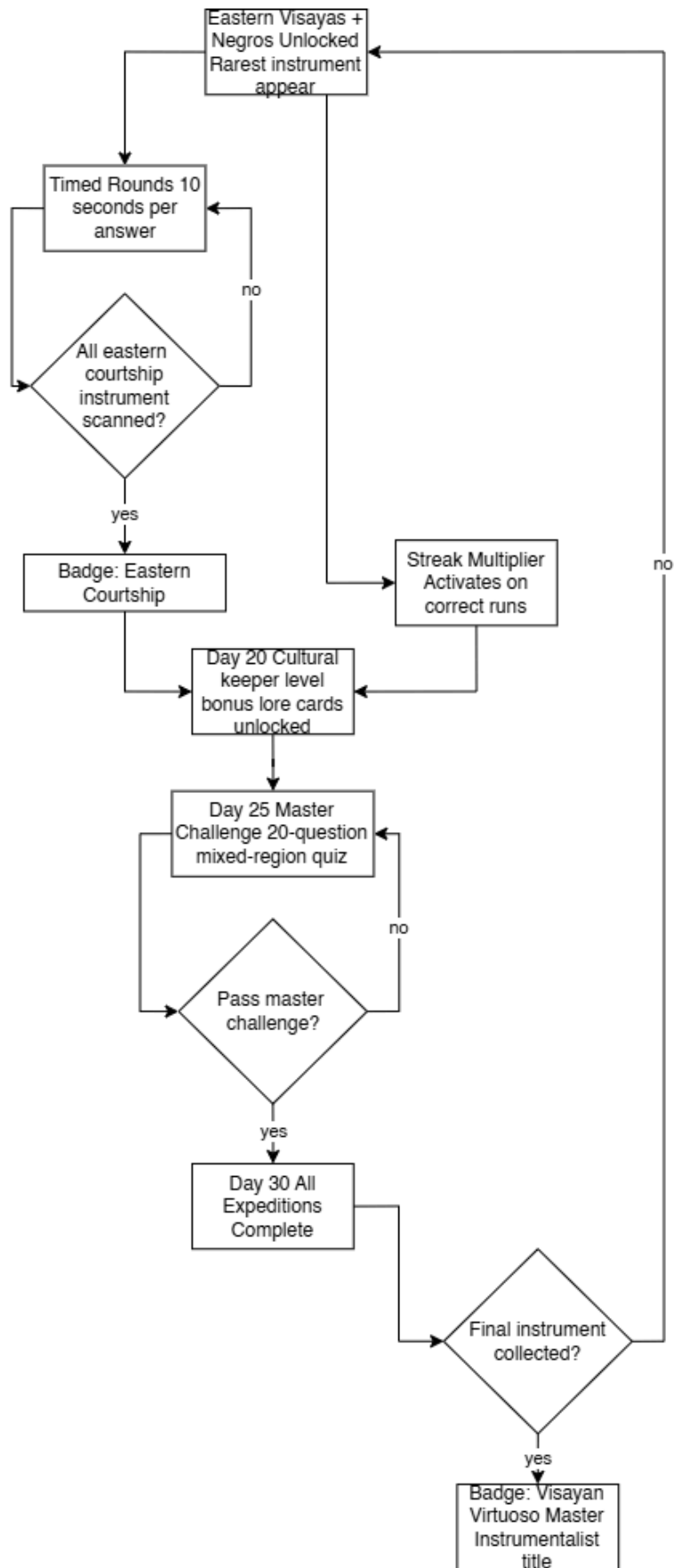
- **Environmental Focus:** The active gameplay background dims into a semi-transparent, deep **Near Black (#0f0c0c)** overlay, eliminating distractions and pulling total player focus to the center of the screen.
- **Card Material & Lighting:** The card itself is forged from a sleek **Dark Slate (#2a2d43)**, bordered by a metallic **Slate Gray (#888ea1)** rim. As the player drags or swipes to rotate the card on its X and Y axes, a dynamic specular highlight sweeps across the surface, catching the light and giving it a premium, tactile feel.
- **Rotational Glow:** While the user manipulates the card, the edges emit a reactive, pulsing **Crimson (#da2d46)** backlight. The intensity of this glow scales with the speed of the user's rotation, rewarding physical interaction.
- **Typography & Lore:** The front of the card features the 3D instrument model. On the reverse side (revealed by spinning the card 180 degrees), the instrument's name is displayed in stark **Light Cool Gray (#e0e5ed)**, accompanied by its ancestral lore or gameplay stats set in a soft, readable **Pale Dusty Pink (#f0dde0)**.

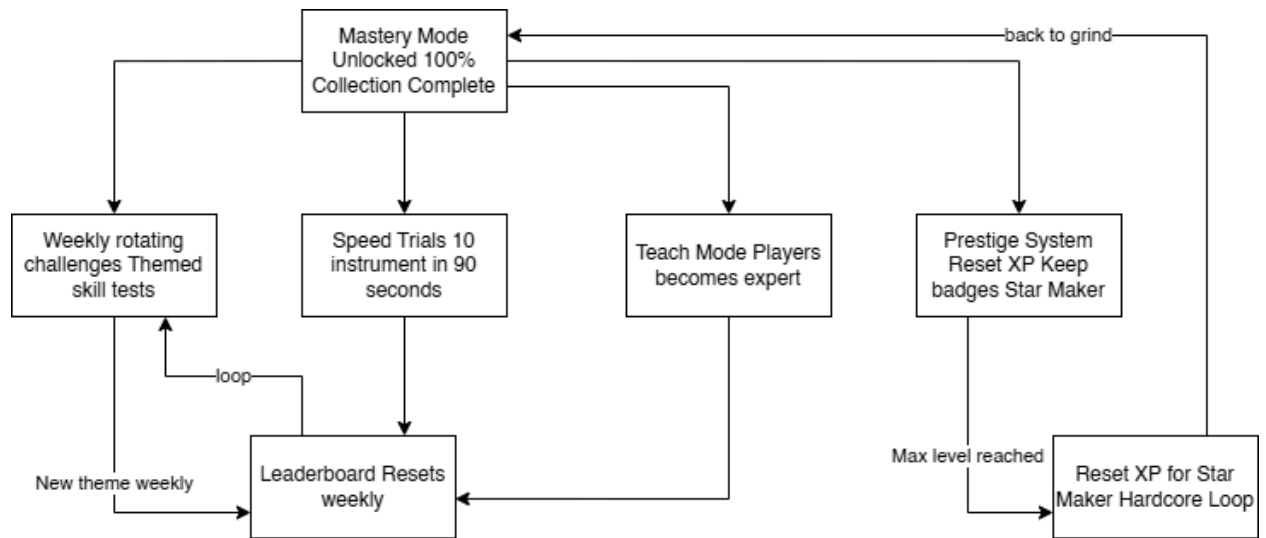
Flowchart











Note:

Location services -> user inputs their location or turns on their location services and the AI will find locations where the user can find scannable instruments like museums etc.

Planned partnerships -> provide us with instrument data